

Does stuff disappear?

Name	Class	Date
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Before anything is weighed Write your prediction down and do not change it. You are not marked on being right. You are marked on whether you can say honestly what happened to your idea.

Part 1 • Predict

A beaker holds 100 g of warm water. Beside it, a cup holds 20 g of salt. Both are on the scale together. The salt gets tipped in and stirred until it vanishes — then the empty cup goes back on the scale.

What will the scale read now?

HEAVIER	LIGHTER	THE SAME
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Circle one. Then write the actual number you expect: _____ g

I think this because...

Part 2 • The dissolving test

Everything stays on the scale. When you tip the salt in, the empty cup goes back on — you are weighing all the stuff you started with.

Mass of beaker + water + cup of salt, BEFORE	
Mass of beaker + solution + empty cup, AFTER	
Difference	

Record to the precision your scales actually give you. If they read whole grams, do not write decimal places — that is an error, not extra care.

Was your prediction right? Write what actually happened.

You cannot see the salt any more. So where is it?

How could you prove it is still there?

Part 3 • The fizzing test

The same reaction, in two different containers. Watch what the scale does differently.

Container A — SEALED	Container B — OPEN
50 mL vinegar in one corner of the freezer bag. 2 g bicarbonate in the other corner. Seal it, keeping them apart. Weigh the whole bag. Tip it so they mix — WITHOUT OPENING IT. Let it finish. Weigh again.	100 mL vinegar in the tall beaker. 6 g bicarbonate in its cup, on the scale beside it. Weigh both together. Tip the bicarbonate in. Put the empty cup back on. Leave it open. Swirl gently, wait five minutes, weigh again.

Do not open the bag Not to look, not to let the pressure out, not at the end. It stays sealed until both readings are written down.

Results

	Mass BEFORE (g)	Mass AFTER (g)	Difference (g)
A — sealed			
B — open			

Explain it

1. Which container changed mass?

2. Both had exactly the same reaction happening inside them. Why did only one change?

3. The bag puffed up. What is inside it — and does it weigh anything? How do you know?

4. Where has the missing mass from container B actually gone?

Part 4 • The whole class

Every group's container B loss goes on the board. Copy them in.

Group	1	2	3	4	5	6
Loss (g)						

Are all the numbers the same? Should they be?

What could make one group's number different from another's? Give two things.

The two big ideas

Nothing disappears

When salt dissolves the mass does not change. The salt is still there, spread through the water in pieces too small to see. Nothing was destroyed, so nothing new was made. That is a REVERSIBLE change.

If it got lighter, something left

When the mass drops in an open container, a substance has escaped — usually as a gas. That gas was not there before, which means something new was made. That is an IRREVERSIBLE change.

Add to the class chart

Last week you listed the evidence that something new has been made. Add the one nobody suggested:

THE MASS CHANGED.

Is this stronger evidence than a colour change? Why?

Push me Predict container B's loss in grams BEFORE you run it, then explain the difference from what you actually got. If you did it again with half the bicarbonate, what would change — and what would stay the same?

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1 • My prediction

100 g of water and 20 g of salt are on the scale together. The salt is stirred in until it vanishes. The empty cup goes back on.

What will the scale read? Circle one.

HEAVIER	LIGHTER	THE SAME
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I think this because _____

2 • What the scale said

Test	Before (g)	After (g)
Salt into water		
Container A — sealed bag		
Container B — open beaker		

3 • Finish these sentences

When the salt dissolved, the mass _____ .
This tells me the salt _____ .
The sealed bag stayed the same because _____

The open beaker got lighter because _____
_____ .

Word bank mass • dissolved • gas • escaped • stayed the same • got lighter • still there • sealed • new substance

Exit tickets — cut into six

Name: _____ A sealed jar of fizzing tablets weighs exactly the same before and after the fizzing stops. <i>A student says: "That proves nothing happened."</i> Are they right? How do you know? _____ _____ _____	Name: _____ A sealed jar of fizzing tablets weighs exactly the same before and after the fizzing stops. <i>A student says: "That proves nothing happened."</i> Are they right? How do you know? _____ _____ _____
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